This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

An 88-dB Max-SFDR 12-bit SAR ADC With Speed-Enhanced ADEC and Dual Registers

***Abstract*—A 12-bit 3-MS/s successive approximation register (SAR) analog-to-digital converter (ADC) was implemented with a modified addition-only digital error correction (ADEC) and a dual-register-based DAC control as speed-enhancement techniques. The proposed speed-enhanced ADEC (SE-ADEC) scheme employs designated capacitors for redundancy-related DAC operation and thus eliminates MUX stages from the DAC control logic. Further speed enhancement is achieved by the dual-register structure that eliminates the entire DAC switching logic. The prototype ADC was fabricated in a 0.35-μm CMOS process utilizing only thick gate transistors with a minimum gate length of 0.5 μm. With a highly linear capacitor DAC design, the prototype ADC achieves 0.38 LSB INL without calibration. The measured spurious-free dynamic range and signal-to-noise and distortion ratio (SNDR) at a low-frequency operation of 250 kS/s are 88 and 68 dB, respectively. At a sample rate of 3 MS/s, the ADC achieves a peak SNDR of 64 dB with a total power dissipation of 1.23 mW under a 2.3-V supply. The $\mathrm{FOM_{Nyq}}$ is a 368-fJ/conversion-step.**

***Index Terms*—Addition-only digital error correction (ADEC), digital error correction, successive approximation register (SAR) ADC.**

I. Introduction

In recent years, successive approximation register (SAR) analog-to-digital converters (ADCs) have been actively researched for various applications by taking full advantage of the merits of low switching power consumption and the enhanced operating speed of advanced low-voltage CMOS processes [1]. However, the majority of commercial products are still fabricated with relatively old technologies with high supply voltage. The ADC in the present work is designed for readout of touch from a 10.1-in touch screen panel, and a 0.35-μm CMOS process was chosen with limited usage of transistors by considering fabrication cost and time: Only thick-gate transistors with a minimum gate length of 0.5 μm were utilized. Even with such an old process, SAR ADCs can still keep a low-power advantage over the other types of ADCs owing to their negligible static power consumption, especially for mobile applications. In order to overcome disadvantages stemming from old technologies, such as low speed and poor-pattern mismatch, this brief proposes several design techniques.

The total capacitance in a high-resolution SAR ADC is large not only due to matching and noise requirements but also due to the design-rule-limited large minimum unit capacitor; thus, the time portion taken by the DAC settling can be considerable. This problem could be alleviated by applying decision redundancies [1]–[3]. The addition-only digital error correction (ADEC) scheme presented in [1] showed a maximum speed enhancement of 37% in a 10-bit ADC compared with the traditional SAR ADC. However, one drawback of the previous ADEC implementation is that the required reswitching logic for the capacitors in MSB segments increases the gate delay in DAC control, which slows down the conversion speed and increases the power consumption. In order to remove the reswitching overhead from the ADEC, this brief introduces a speed-enhanced ADEC (SE-ADEC) technique that employs dedicated capacitors for the redundancy-related DAC control. Furthermore, the proposed dual-register-based DAC control scheme eliminates the entire switching logic for DAC and thus increases the conversion speed.

In order to achieve 12-bit linearity without capacitor calibration, a highly linear capacitor DAC (CDAC) is designed with special care on the minimization of the parasitic-induced capacitance mismatch rather than on the conventional design strategy of the common-centroid layout. The measurement results reveal that the matching between the unit capacitors is good enough for 12-bit linearity even without a common centroid and that the parasitic capacitance mismatch by uneven interconnection patterns can be a more serious source of nonlinearity like that in [1].

The remainder of this brief is organized as follows. The operational principle of the proposed SE-ADEC technique is explained in Section II. In Section III, the circuit implementation of a prototype 12-bit SE-ADEC SAR ADC is presented with the DAC structure and the proposed dual-register-based DAC control. Section IV demonstrates the experimental results, and Section V concludes this brief.

II. SE-ADEC Method

Fig. 1 compares two 4-bit SAR ADCs: Fig. 1(a) with the original ADEC [1] and Fig. 1(b) with the proposed SE-ADEC. The SAR ADC with the original ADEC has a binary-weighted DAC that is virtually divided into two sub-DACs. While the scheme retains the binary weighted DAC, the

Manuscript received March 19, 2013; revised May 1, 2013; accepted June 8, 2013. This work was supported in part by the Center for Integrated Smart Sensors as Global Frontier Project (CISS-2012M3A6A6054202) and the National Research Foundation (NRF) National Research Foundation of Korea (2012R1A2A2A01047062) funded by the Korean government (Ministry of Science, ICT & Future Planning). The CAD tools were supported by IDEC of KAIST. This brief was recommended by Associate Editor X. Jiang.

S.-Y. Baek and J.-K. Lee were with the Korea Advanced Institute of Science and Technology (KAIST), Daejeon 305-732, Korea. They are now with Samsung Electronics Company, Yongin 446-711, Korea.

S.-T. Ryu is with the Department of Electrical Engineering, KAIST, Daejeon 305-732, Korea (e-mail: stryu@ kaist.ac.kr).

Color versions of one or more of the figures in this brief are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TCSII.2013.2268434

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

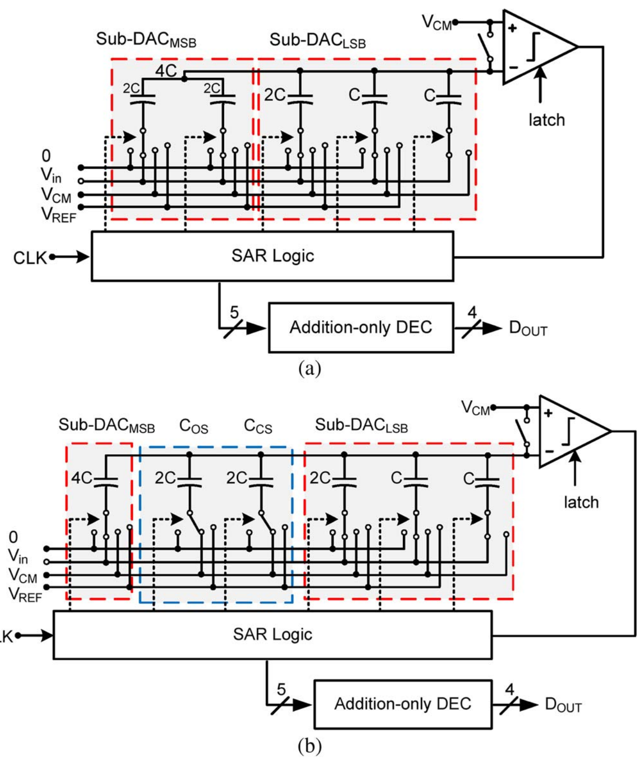
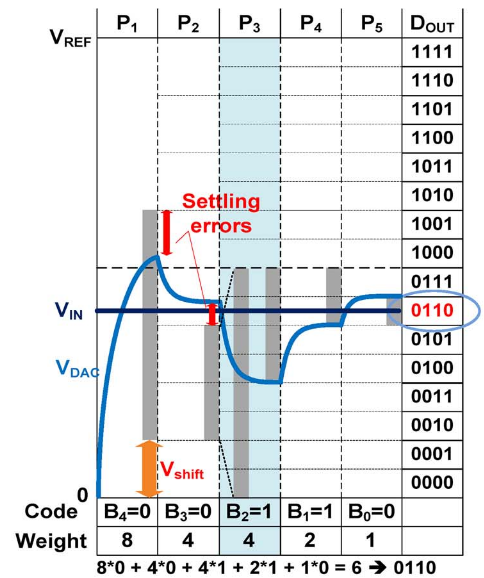


Fig. 1. SAR ADCs with (a) original ADEC and (b) SE-ADEC.

capacitor reswitchings in the sub-DAC$_{MSB}$ for decision redundancy requires additional MUX in the DAC control logic, which degrades the speed. In order to eliminate this drawback while keeping the same principle of ADEC, the capacitor 4C in sub-DAC$_{MSB}$ remains undivided, and instead, the dedicated capacitors, C_{OS} and C_C, are added in the proposed SE-ADEC. While other capacitors in sub-DACs work as a normal binary weighted CDAC, C_{OS} and C_C work only for the decision threshold shift for ADEC with an amount of half the LSB of the sub-DAC$_{MSB}$.

The CDAC operation of the proposed SE-ADEC scheme in Fig. 1(b) is explained with Fig. 2. Fig. 2(a) shows the DAC waveform in each decision phase, and the bottom-plate connections of the capacitors are shown in Fig. 2(b) for $V_{IN} = 13/32\ V_{REF}$. Note that the DAC voltage (V_{DAC}) and the sampled input (V_{IN}) are drawn separately for convenience. During the input sampling period (not shown), all capacitors for a conventional binary DAC sample the input signal while C_{OS} and C_C sample V_{CM} for zero charge in them. This zero charge sampling in C_{OS} and C_C reduces the signal gain at the comparator input but then the signal gain becomes the same as that of the DAC, eliminating the decision gain error. For the MSB decision at P_1, all capacitors are connected to V_{CM}, as in the conventional straightforward decision scheme [1], except for C_{OS}, which is connected to V_{REF} for the decision threshold shift for ADEC. Since the MSB (B_4) is zero, in the following phase of P_2, the MSB capacitor (4C) is connected to zero as in a normal SAR ADC. In P_3, the redundant decision phase with over-range, the following operations are conducted: Unlike the basic ADEC, reswitching of the MSB capacitor (4C) and C_{OS} are not necessary. When $B_3 = 0$, the same amount of decision offset added by C_{OS} is removed by connecting C_C to zero. When $B_3 = 1$, the same amount of the offset by C_{OS} is then added by connecting C_C to V_{REF}. This dedicated capacitor usage for decision offset control makes the SAR ADC with the SE-ADEC faster than that with the original ADEC by eliminating the MUX required for the DAC reswitching. As Fig. 2(a) shows, the decision principle is exactly the same as that of the original ADEC, and the ADC achieves a correct result even with settling errors in P_1 and P_2.



(a)

V_{in} = 7/32V_{REF}	4C	Cos	Cc	2C	C	C	Code
Sample	V_{in}	V_{CM}	V_{CM}	V_{in}	V_{in}	V_{in}	
P_1	V_{CM}	V_{REF}	V_{CM}	V_{CM}	V_{CM}	V_{CM}	B_4=0
P_2	0	V_{REF}	V_{CM}	V_{CM}	V_{CM}	V_{CM}	B_3=0
P_3	0	V_{REF}	0	V_{CM}	V_{CM}	V_{CM}	B_2=1
P_4	0	V_{REF}	0	V_{REF}	V_{CM}	V_{CM}	B_1=1
P_5	0	V_{REF}	0	V_{REF}	V_{REF}	V_{CM}	B_0=0

(b)

Fig. 2. SE-ADEC operation of the SAR ADC shown in Fig. 1(b). (a) DAC waveform in each decision phase and (b) capacitor bottom-plate connection.

Referring to Figs. 1 and 2, the C_{OS} and C_C may make the proposed SE-ADEC scheme look inefficient in terms of hardware complexity and power consumption because of the MSB-capacitor comparable values. However, in real-circuit implementation, the decision over-range will be given much smaller C_{OS} and C_C than the MSB capacitor. As for the power consumption, it can be found that SE-ADEC consumes less power than the original ADEC does. Even though the original ADEC does not have additional capacitors, the same charging and discharging by C_{OS} and C_C are conducted by the 2Cs in the sub-DAC$_{LSB}$ and sub-DAC$_{MSB}$. What is worse with the original ADEC is that the capacitors in sub-DAC$_{MSB}$ reswitches when the 2C in sub-DAC$_{LSB}$ returns to V_{CM}[1] and consumes additional power in charging and discharging the capacitors and in operating the reswitching logic (MUX). On the other hand, this reswitching-related power overhead is removed in the SE-ADEC.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

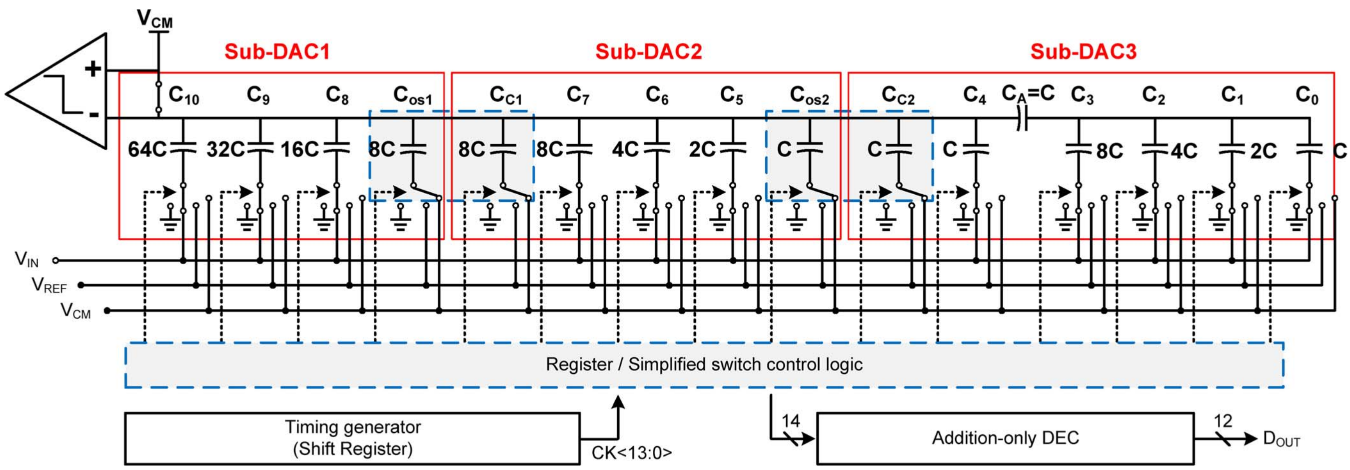


Fig. 3. Circuit diagram of the 12-bit SAR ADC with the proposed SE-ADEC scheme (single-end version).

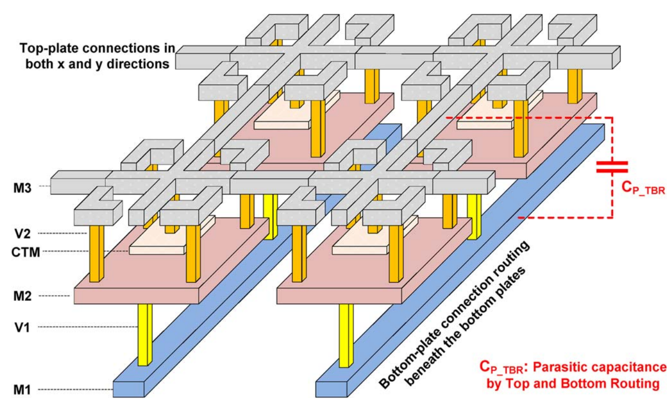


Fig. 4. CDAC structure and the unit-capacitor structure for minimization of routing-induced parasitic capacitance.

III. Circuit Implementation

A. 12-bit Linear SAR ADC With Three-Step SE-ADEC

Fig. 3 shows the structure of the 12-bit SAR ADC with the SE-ADEC scheme. Due to the limitation of the given process, MiM capacitors were used for the CDAC. The attenuation-capacitor-based segmented-DAC structure was chosen to reduce the input capacitance and chip area. The unit capacitor C was determined to be 25 fF because it was the minimum allowed by the process while showing acceptable matching property for the design target. The physical location of the attenuation capacitor $\mathrm{C_A}$ was determined by considering its top-plate parasitic effect on the DAC linearity, and 4-bit LSBs were assigned after $\mathrm{C_A}$. As Fig. 3 shows, the CDAC is virtually divided into three sub-DACs by referring to the result in [1] considering resolution similarity, and additional capacitor pairs $(\mathrm{C_{OS1-2}}, \mathrm{C_{C1-2}})$ are added between the sub-DACs for the proposed SE-ADEC operation. These capacitors increase the total capacitance by 13%. After 14 successive decisions including redundancies, the final 12-bit output code is achieved via the ADEC logic.

Guaranteeing a 12-bit linearity from a CDAC without mismatch calibration is known to be a nontrivial task because of random and systematic mismatches between a large number of unit capacitors. A common centroid placement of unit capacitors has often been recommended to reduce a geometry-dependent capacitor mismatch. However, the previous work from [1] showed that the capacitor mismatches in DAC are rather dominated by the complicated routings of top-plate metal connection. The complicated top-plate interconnections for the common-centroid placement make the parasitic capacitance associated with each unit capacitor uneven in conjunction with the capacitor bottom plate and result in a systematic capacitance mismatch. On the other hand, the random mismatch of devices is known to be a weak function of the distance between devices [4]. Thus, this study focuses on how to ensure that every unit capacitor has identical interconnection-induced parasitic capacitance while trying to minimize the parasitic amount. In order to minimize the fringing flux from the top layer of one capacitor to the bottom layer of other capacitors, each capacitor's top plate and the routing metal from it are surrounded by the metal layers connected to the bottom plate of the capacitor itself, as shown in Fig. 4. In addition, the interconnection metal for the bottom plates of the capacitors runs beneath the bottom plates to minimize the coupling capacitance with the common top plate. In order to ensure identical parasitic capacitance $(\mathrm{C_{P\_TBR}})$ from the routing interconnections for every unit capacitor, the bottom-plate routing of a unit capacitor was drawn in such a way that it overlaps with the top-plate routing in two directions out of the cross-shaped four directions. Special care is needed in laying out the attenuation capacitor $\mathrm{C_A}$ because it is the only floating capacitor in the CDAC, and different routing environment produces different parasitic $(\mathrm{C_A} \neq \mathrm{C})$. This problem was resolved with the aid of postlayout parasitic extraction and by adjusting the length of the interconnection metal on $\mathrm{C_A}$.

B. Dual-Register-Based Fast DAC Control

As mentioned in Section II, the logic delay in the decision loop is one of the major sources of speed limitation in a SAR ADC. In this design, DAC switching logic is not just simplified by the SE-ADEC but rather completely eliminated by the proposed dual-register-based direct DAC control scheme. This technique is similar to the register-to-DAC direct-control

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

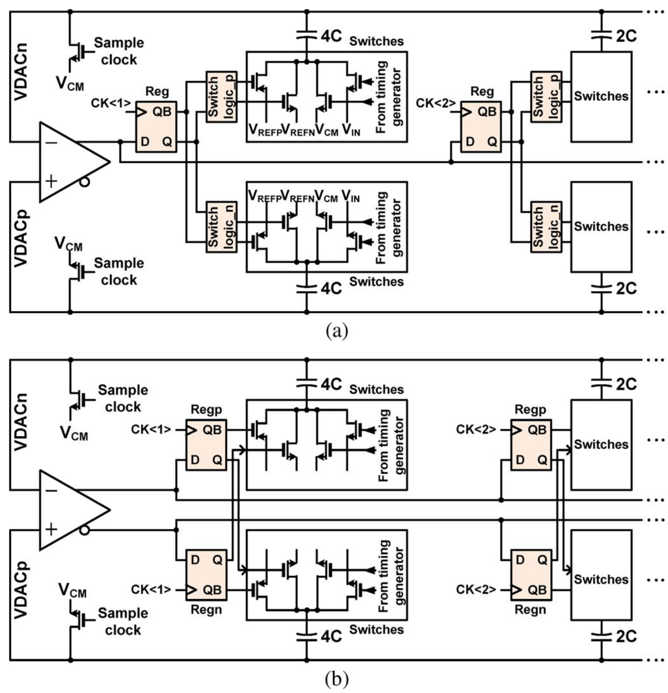


Fig. 5. SAR ADC with (a) conventional single-register-based DAC control and (b) the proposed dual-register-based direct DAC control.

technique used in the 2-bit/cycle SAR ADC [5] in that they both eliminate DAC control logic except that the design in [5] uses a new register and does not use $\mathrm{V_{CM}}$. Fig. 5 compares a conventional single-register-based DAC control scheme [Fig. 5(a)], and the proposed dual-register-based scheme [Fig. 5(b)]. For simplicity and consistency with Fig. 1, only the circuit configuration related to MSB capacitors is shown for the 4-bit example. Instead of using a single register and capacitor switching logic for 1-bit capacitor control as in the conventional design, in the proposed design, dual registers are used (Regp and Regn) without additional switch logic. The output Q of the P-side register (Regp) is connected to the $\mathrm{V_{REFN}}$ switch in the P-side, and the QB of the same register is connected to the $\mathrm{V_{REFP}}$ switch in the N-side. Q and QB of the N-side register (Regn) are connected to the $\mathrm{V_{REFN}}$ switch of the N-side and the $\mathrm{V_{REFP}}$ switch of the P-side, respectively. Since both outputs of the comparator are reset to zero in the reset phase (when conversion begins) in this design, all the registers have $\mathrm{Q} = 0$ and $\mathrm{QB} = 1$. This makes all the switches connected to $\mathrm{V_{REFP}}$ and $\mathrm{V_{REFN}}$ turn off. By the dual registers operating in sequence by CK$\langle \mathrm{i} \rangle$, the capacitors in both the P-side and N-side are directly controlled simultaneously in turn without the aid of any switching logic. Note that the $\mathrm{V_{IN}}$ and $\mathrm{V_{CM}}$ switches are controlled by the global timing generator shown in Fig. 3. Owing to the dual-register-based DAC control scheme, the proposed SE-ADEC SAR ADC saves two cascaded MUXes from the DAC control path compared with the original ADEC.

A conventional dynamic latch was used for a comparator in this design without a preamplifier, similar to that in [1]. The comparator noise design was conducted based on the transient noise simulation. Owing to the high supply voltage and high output resistance of 0.5-μm transistor, the latch provides high gain when it starts to form the positive feedback loop, and the noise performance could be achieved with no special technique. The simulated input referred noise voltage corresponds to 1/4 LSB.

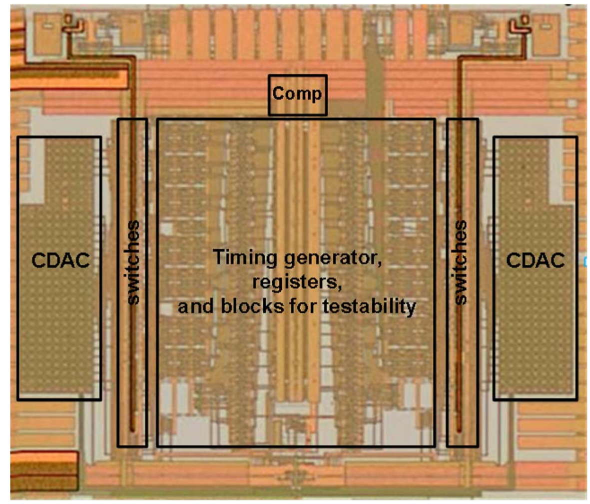


Fig. 6. Chip photo.

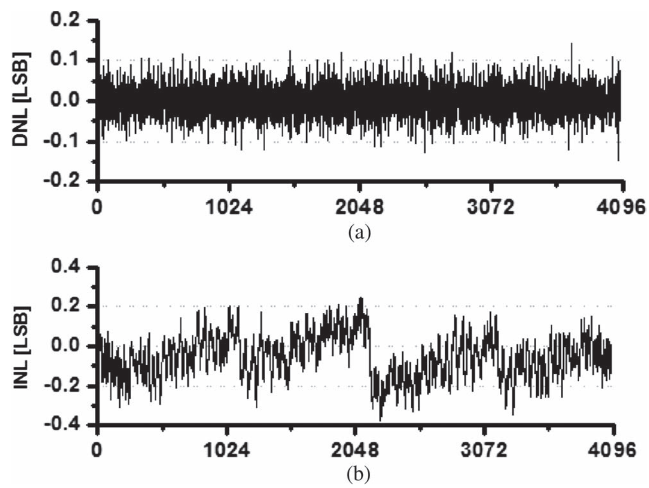


Fig. 7. DNL and INL.

IV. Measurement Results

A prototype ADC was fabricated in a 1P4M 0.35-μm process and only 0.5-μm CMOS transistors were utilized. A chip photo is shown in Fig. 6. The core size is $560 \times 615\ \mu\mathrm{m}^2$ (39% of the digital portion is occupied by test circuits). The chip operates under a 2.3-V supply voltage. In order to test the capacitor-mismatch limited linearity, differential non-linearity (DNL) and integral non-linearity (INL) were measured with a high-precision 2.5-kHz sinusoidal signal from an audio signal source at a 250-kHz sample rate. The achieved DNL and INL were $-0.14{-}0.14$ LSB and $-0.38{-}0.25$ LSB, respectively (Fig. 7). This demonstrates the excellent capacitor matching property of the proposed capacitor structure and the routing strategy. Fig. 8 shows the fast Fourier transform (FFT) result with the same measurement setup as employed for the static linearity. The measured spurious-free dynamic range (SFDR) is 88 dB, and the signal-to-noise and distortion ratio (SNDR)

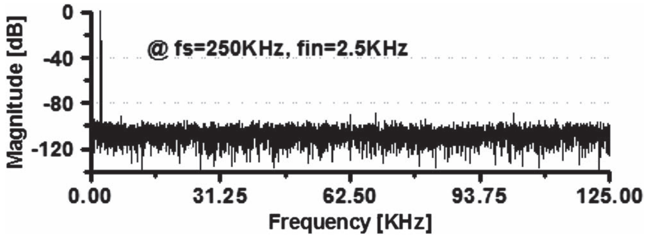


Fig. 8. FFT result at 250 kS/s with an input frequency of 2.5 kHz.

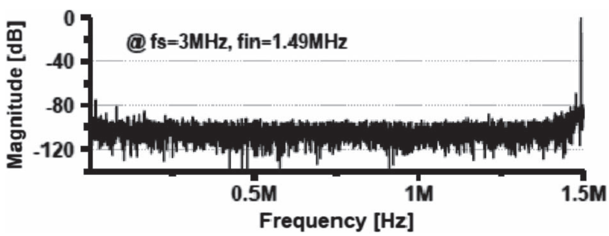


Fig. 9. FFT result at 3 MS/s with an input frequency of 1.49 MHz.

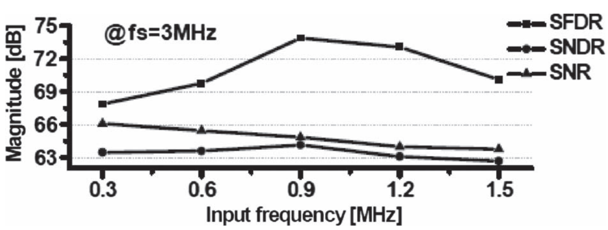


Fig. 10. Dynamic performances at 3-MS/s conversion with various input frequencies.

is 68 dB (11 ENOB) at the input of -1 dBFS. The power consumption at a 250-kS/s conversion rate was 107 μW, resulting in a 209-fJ/conversion-step of low-frequency figure of merit (FOM). At higher frequency input signals above 300 kHz, the noise and linearity performances were limited by the signal source and the filter we used. Fig. 9 shows the FFT result measured at 3 MS/s with a Nyquist input of -1 dBFS. The SFDR and SNDR are 70.1 and 62.7 dB, respectively. The total power consumption at a 3-MS/s conversion is 1.23 mW, and the FOM at a Nyquist rate is 368 fJ/conversion-step. The dissipated power by the comparator, charging and discharging the DAC, and the digital logic are 18 μW, 68 μW, and 1.14 mW, respectively. Unlike in advanced CMOS processes, the power consumed by the digital block accounts for 93% of the total power consumption. Fig. 10 shows the dynamic performances of the prototype ADC for various input frequencies at 3-MS/s conversion. The peak SNDR was achieved as 64 dB at 0.9-MHz input, which is limited by the low-pass filter we used in the test.

It is not easy to compare the performance of the prototype design with designs fabricated in relatively recent processes because of a large technology gap. Therefore, a design with similar specifications to this work was found in the literature for convenient comparison. A 1.0-V 12-bit SAR ADC fabricated in 0.13 μm reported in [6] demonstrates a performance tested at 3-MS/s conversion rate. Table I shows the comparison. The INL of this work is eight times lower than that in [6] despite the old technology. The prototype ADC in this study consumes only 65% of the power of that in [6] at 3 MS/s while operating under a supply voltage that is 2.3 times higher than that in [6]. The chip area of the prototype is almost half of that in [6].

TABLE I
PERFORMANCE COMPARISON WITH SIMILAR SPECIFICATION ADC

	Kang, VLSI 2009 [6]		**This work**	
Process	0.13 μm CMOS		0.5 μm CMOS in a 0.35 μm process	
Supply	1.0 V		2.3 V	
Resolution	12 b		12 b	
DNL	0.8 LSB		0.14 LSB	
INL	3.0 LSB		0.38 LSB	
Sample rate	11 MS/s	**3 MS/s**	**3 MS/s**	0.25 MS/s
Power	3.57 mW	**1.88 mW**	**1.23 mW**	0.107 mW
SFDR	70 dB	**72 dB**	**70.1 dB**	88 dB
SNDR	62 dB	**65.3 dB**	**62.7 dB**	68 dB
FOM	311 fJ/c-s	**400 fJ/c-s**	**368 fJ/c-s**	209 fJ/c-s
Area	0.7 mm^2		0.34 mm^2	

V. CONCLUSION

This brief has presented various design techniques for speed enhancement and linearity improvement for SAR ADCs. The speed-enhancing technique for the ADEC removed the capacitor reswitching operation by modifying the DAC structure and capacitor switching algorithm. In addition, the dual-register-based direct DAC control method eliminates the entire DAC switching logic. As a result, the prototype ADC could save two cascaded MUXes from the DAC control path compared with the original ADEC. In addition, the proposed CDAC layout strategy has showed an excellent linearity. Owing to all the design schemes proposed in this brief, the prototype ADC has showed a competitive performance compared with an ADC designed in a relatively advanced process.

ACKNOWLEDGMENT

The authors would like to thank the IC Design Education Center of KAIST for the CAD tools.

REFERENCES

[1] S.-H. Cho, C.-K. Lee, J.-K. Kwon, and S.-T. Ryu, "A 550-μW 10-b 40-MS/s SAR ADC with multistep addition-only digital error correction," *IEEE J. Solid-State Circuits*, vol. 46, no. 8, pp. 1881–1892, Aug. 2011.

[2] C. C. Liu, S.-J. Chang, G.-Y. Huang, Y.-Z. Lin, C.-M. Huang, C.-H. Huang, L. Bu, and C.-C. Tsai, "A 10 b 100 MS/s 1.13 mW SAR ADC with binary-scaled error compensation," in *Proc. IEEE ISSCC Dig. Tech. Papers*, Feb. 2010, pp. 386–387.

[3] F. Kuttner, "A 1.2-V 10-b 20-MSample/s non-binary successive approximation ADC in 0.13-μm CMOS," in *Proc. IEEE ISSCC Dig. Tech. Papers*, 2002, pp. 176–177.

[4] M. J. M. Pelgrom, A. C. J. Duinmaijer, and A. P. G. Welbers, "Matching Properties of MOS Transistors," *IEEE J. Solid-State Circuits*, vol. 24, no. 5, pp. 1433–1440, Oct. 1989.

[5] H.-K. Hong, W. Kim , S.-J. Park, M. Choi, H.-J. Park, and S.-T. Ryu, "A 7b 1 GS/s 7.2 mW nonbinary 2b/cycle SAR ADC with register-to-DAC direct control," in *Proc. IEEE CICC*, Sep. 2012, pp. 1–4.

[6] J. J. Kang and M. P. Flynn, "A 12b 11 MS/s successive approximation ADC with two comparators in 0.13 μm CMOS," in *IEEE Symp. VLSI Circuits*, Jun. 2009, pp. 240–241.